INCH-POUND
FF-W-92C
June 7, 2021
SUPERSEDING
FF-W-92B
May 9, 1974

FEDERAL SPECIFICATION
WASHER, FLAT (PLAIN)

Reactivated after 7 June 2021 and may be used for new and existing designs and acquisitions.

The General Services Administration has authorized the use of this federal specification by all federal agencies, some states and localities.

1 SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plain, flat washers.

1.2 Classification. Flat washers covered by this specification shall be of the following types, grades and classes, as specified (see 6.2).

Type:

A - Round washer
B - Square washer

Grade:

I - General assembly purpose washer
II - Aeronautical assembly purpose washer

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: U.S. Army CCDC, Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-IES, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or usarmy.detroit.devcom-gvsc.mbx.standardization@mail.mil.

FSC 5310

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

Class:

A - Carbon steel washer
B - Corrosion-resisting steel washer
C - Aluminum washer
D - Aluminum alloy washer
E - Brass washer
F - Copper washer
G - Nickel-copper alloy
H - Plastic (nylon) washer

2 APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FEDERAL SPECIFICATIONS

QQ-N-281	-	Nickel-Copper Alley Bar, Rod, Plate, Sheet, Strip, Wire, Forgings and Structural and Special Shaped Sections

FEDERAL STANDARDS

FED-STD-123	-	Marking for Shipment (Civil Agencies)

MILITARY SPECIFICATIONS

MIL-DTL-13924	-	Coating, Oxide, Black, For Ferrous Metals
MIL-DTL-16232	-	Phosphate Coatings, Heavy, Manganese or Zinc Base
MIL-F-495	-	Finish, Chemical, Black, For Copper Alloys
MIL-PRF-8625	-	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-PRF-32647	-	Zinc-Nickel Electroplating for Fasteners

MILITARY STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-2073-1	-	Standard Practice for Military Packaging

(Copies of specifications and standards are available for download from the ASSIST database at https://quicksearch.dla.mil/.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Private sector and civilian agencies may purchase copies of these voluntary standards from the American National Standards Institute, 1430 Broadway, New York, NY, 10018, help@asq.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.21.1	-	Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

(Private sector and civilian agencies may purchase copies of these voluntary standards from the American Society of Mechanical Engineers, Two Park Avenue, New York, NY 10016-5990, CustomerCare@asme.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) INTERNATIONAL

ASTM A153/A153M	-	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
ASTM-A342/A342M	-	Standard Test Methods for Permeability of Weakly Magnetic Materials
ASTM-A967/A967M	-	Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts
ASTM-B36/B36M	-	Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar
ASTM-B121/B121M	-	Standard Specification for Leaded Brass Plate, Sheet, Strip, and Rolled Bar
ASTM-B152/B152M	-	Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar
ASTM B209	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM-B339	-	Standard Specification for Pig Tin
ASTM-B545	-	Standard Specification for Electrodeposited Coatings of Tin
ASTM-B633	-	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

ASTM-D3951	-	Standard Practice for Commercial Packaging
ASTM-D4066	-	Standard Classification System for Nylon Injection and Extrusion Materials (PA)

(Private sector and civilian agencies may purchase copies of these voluntary standards from the American Society Testing and Materials, Race Street, Philadelphia., PA, 19103, https://www.astm.org/Standard/.)

AEROSPACE INDUSTRIES ASSOCIATION/NATIONAL AEROSPACE STANDARD (AIA/NAS)

AIA/NAS NAS1149	-	Washer, Flat

(Private sector and civilian agencies may purchase copies of these voluntary standards from the Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700 Arlington, VA 22209-3928, https://global.ihs.com/home_page_aia.cfm?&csf=AIA.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE AMS2700	-	Passivation of Corrosion Resistant Steels
SAE AIR4127	-	Steel: Chemical Composition and Hardenability
SAE-AMS-C-81562	-	Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited)
SAE-AMS-QQ-A-250	-	Aluminum and Aluminum Alloy, Plate and Sheet General Specification for
SAE-AMS-QQ-N-290	-	Nickel Plating (Electrodeposited)
SAE-AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)

(Private sector and civilian agencies may purchase copies of these voluntary standards from the Society of Automotive Engineers International, 400 Commonwealth Drive, Warrendale, PA 15096, https://www.sae.org/standards/.)

3 REQUIREMENTS

3.1 Material.

3.1.1 Class A carbon steel. Carbon steel washers shall be low carbon steel in accordance with SAE AIR4127.

3.1.2 Class B corrosion-resistant steel. Corrosion-resisting steel washers shall be type 301, 302, 304, 305 or 316 in accordance with SAE AIR4127.

3.1.2.1 Magnetic permeability. Magnetic permeability shall be 2.0 maximum (air = 1.0) for a field strength of H = 200 oersteds when using a magnetic indicator per ASTM-A342/A342M.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.1.3 Class C – aluminum. Aluminum washers shall be in accordance with ASTM B209, temper H18 or H28.

3.1.4 Class D - aluminum alloy. Aluminum alloy washers shall be in accordance with SAE-AMS-QQ-A-250, temper T4.

3.1.5 Class E – brass. Brass washers shall be in accordance with ASTM-B36/B36M and ASTM-B121/B121M, composition 11, temper half-hard.

3.1.6 Class F – copper. Copper washers shall 'be in accordance with ASTM-B152/B152M, temper cold-rolled, hard.

3.1.7 Class G - nickel-copper alloy. Nickel-Copper alloy washers shall be in accordance with QQ-N-281, Class A, cold rolled, solution annealed and aged hardened.

3.1.8 Class H - plastic (nylon). Plastic (nylon) washers shall be in accordance with ASTM-D4066, Type I, natural color.

3.2 Protective finish. Unless otherwise specified (see 6.2), washers shall be furnished plain, with no protective finish. When finishes are specified, they shall be in accordance with the following paragraphs.

3.2.1 Class A - carbon steel washers.

3.2.1.1 Cadmium plating (Obsolete for New Design, see 6.3). Cadmium /1 plate typing shall be in accordance with SAE-AMS-QQ-P-416; Type II, Class 3 or SAE-AMS-C-81562, II, Class 3.

/1 Cadmium is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K16 is the preferred method.

3.2.1.2 Zinc-Nickel plating. Zinc-Nickel plating shall be in accordance with MIL-PRF-32647, Type K16.

3.2.1.3 Zinc coating. Zinc coating shall be in accordance with ASTM-B633 /2, Type II or SAE-AMS-C-81562, type II. The thickness on washers for Bolt sizes thru 0.250 inch shall be Class 3 and on washers for bolt sizes over 0.250 inch shall be Class 2.

/2 Type II passivation per ASTM B633 contains hexavalent chromium which is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc with hex free passivation per ASTM F1941/F1941M Fe/Zn 12 or Zinc-Nickel plate per MIL-PRF-32647 Type K20 are the preferred methods.

3.2.1.4 Galvanized zinc. Hot-dip galvanizing shall be in accordance with ASTM A153/A153M. The thickness shall be 0.00045 inch minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.2.1.5 Phosphate coating. Phosphate coating shall be in accordance with MIL-DTL-16232, Type Z, Class 2.

3.2.2 Class B - corrosion-resisting steel washer.

3.2.2.1 Passivation. Passivation treatment shall be in accordance with ASTM-A967/A967M or SAE AMS2700 1/.

/1 Passivation per SAE AMS2700 contains sodium dichromate which is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc with hex free passivation per ASTM F1941/F1941M Fe/Zn 12 or Zinc-Nickel plate per MIL-PRF-32647 Type K20 are the preferred methods.

3.2.2.2 Black oxide. Black oxide coating shall be in accordance with MIL-DTL-13924, Class 4.

3.2.3 Classes C and D - aluminum and aluminum alloy washers.

3.2.3.1 Anodizing. Anodizing shall be in accordance with MIL-DTL-5541, Type II, Class 1A.

3.2.4 Classes E and F - Brass and copper washers.

3.2.4.1 Tin plating. Tin plating shall be in accordance with ASTM-B545, ASTM-B339.

3.2.4.2 Nickel plating. Nickel plating shall be in accordance with SAE-AMS-QQ-N-290, Class l, Grade F.

3.2.4.3 Black chemical. Black chemical finish shall be in accordance with MIL-F-495.

3.3 Dimensions.

3.3.1 Type A, round washers.

3.3.1.1 Grade I. Grade I, general assembly purpose washers shall be in accordance with the dimensions in the applicable military specification and ASME B18.21.1.

3.3.1.2 Grade II. Grade II, aeronautical assembly purpose washers shall be in accordance with the dimensions in AIA/NAS NAS1149.

3.3.1.3 Washer faces. Washer faces shall be parallel within 0.002 inch. Washer faces shall be flat within:

0.007 inch for bolt sizes thru 0.500 inch.
0.010 inch for bolt sizes 0.5625 thru 1.250 inches.
0.015 inch for bolts sizes over 1.250 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.3.1.4 Concentricity. The diameter and the outside diameter shall be concentric within the tolerance of the inside diameter.

3.3.2 Type B square washers. Square washers shall be in accordance with the dimensions in Table I.

3.4 Workmanship. Washers shall be flat and smooth and free from loose scale, sharp edges and all other defects which may affect their serviceability.

Table I. Type B, grade I square washers.

Nom. bolt size	Inside diameter		Outside width		Thickness	
	Min.	Max.	Min.	Max.	Min.	Max.
.375	.438	.469	1.50	1.56	.125	1.56
.4735	.500	.531	1.75	1.81	.125	1.56
.500	.562	.594	2.00	2.06	.188	.219
.625	.719	.750	2.25	2.31	.250	.281
.750	.844	.785	2.50	2.56	.250	.281
.875	.969	1.000	3.00	3.06	.250	.281
1.000	1.094	1.125	3.50	3.56	.375	.406
1.125	1.250	1.312	4.00	4.12	.375	.406
1.250	1.375	1.429	4.50	4.62	.375	.406
1.375	1.500	1.562	5.00	5.12	.375	.406
1.500	1.625	1.688	6.00	6.12	.375	.406
1.750	1.875	1.938	6.50	6.62	.375	.406
2.000	2.125	2.188	7.00	7.12	.375	.406

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in this contract or order, the supplier may use his awn or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this and cited specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all washers of the same type, grade, class, size and protective finish, produced under the same conditions and offered for acceptance at one time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

4.3 Sampling.

4.3.1 Sampling for examination. A random sample of washers shall be selected from each lot in accordance with ANSI/ASQ Z1.4, inspection level S-2. The Acceptable Quality Level (AQL) defective shall be 4.0 percent defective.

4.3.2 Sampling for tests. The sampling method shall be in accordance with MIL- STD-1916 or ASQ H1331, table 1 or a comparable zero based sampling plan unless otherwise specified by the contact. If the applicable drawing, specification, standard, or quality assurance provision (QAP) specifies Critical, Major and/or Minor attributes, they shall be assigned Verification Levels of VII, IV and II or AQLs of 0.1, 1.0 and 4.0 respectively. Unspecified attributes shall be considered as Major unless sampling plans are specified in applicable documents. For MIL-STD-1916 the manufacturer may use the attribute or variable inspection method at their option. MIL-STD-105/ASQ Z1.4 may be used to set sample lot size but acceptance is at zero non-conformances in the sample lot unless otherwise specified in the contract

4.4 Examination. Samples selected in accordance with 4.3.1 shall visually examined for material (3.1), protective finish (3.2), dimensions (3.3), and workmanship (3.4). Corrosion resistant steel washers shall be checked for magnetic permeability (3.1.2.1). Any washer containing one more defects shall be rejected, and if the number of defective washers exceed the acceptance number, the lot represented by the sample shall be rejected.

4.5 Tests.

4.5.1 Protective finish. Tests for protective finishes shall be in accordance with the applicable specification (see 3.2).

4.5.2 Preparation for delivery. Tests for preparation for delivery shall be in accordance with ASTM-D3951 or MIL-STD-2073-1 to verify conformance with Section 5.

5 PACKAGING

5.1 Packaging and packing. Packaging shall be Level A or C and packing shall be Level A, B, or C in accordance with ASTM-D3951 or MIL-STD-2073-1, as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to marking required by the contract or order, all interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.2.2 Military agencies. In addition to marking required by the contract or order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

6 NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. Flat washers are intended for automotive, aircraft and general bolt, nut and screw application. Their purpose is to provide an increased bearing surface and to prevent galling of the base material.

6.2 Ordering data. Purchasers should select the offered options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Type, grade and class (see 1.2).
c. Protective finish, if required (see 3.2).
d. Dimensions (see 3.3)
e. Selection of applicable levels of packaging and packing (see 5.1).
f. Applicable MS or AN part number.

6.3 Military Procurement. Zinc-Nickel plated fasteners should be used for replacement parts. Items procured under this specification are for military use. It is recommended the design as specified herein be evaluated for use. Cadmium or Hexavalent-Chromium should only be used in high temperature applications.

6.4 Subject term (key word) listing.

Protective finish
Cadmium
Hexavalent-Chromium
Concentricity
Zinc-Nickel

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

6.6 Supersession data. This specification includes the requirements of FF-W-92B, base document dated May 9, 1974, and incorporates Amendment 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

Military Custodians:
Army - AT
Navy - OS
Air Force - 71
DLA – IS

Reviewer Activities
Army - AV, CE
Navy - AS, MC, SH, YD

Preparing Activity:
Army - AT

(Project No. 5310-2021-019)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.